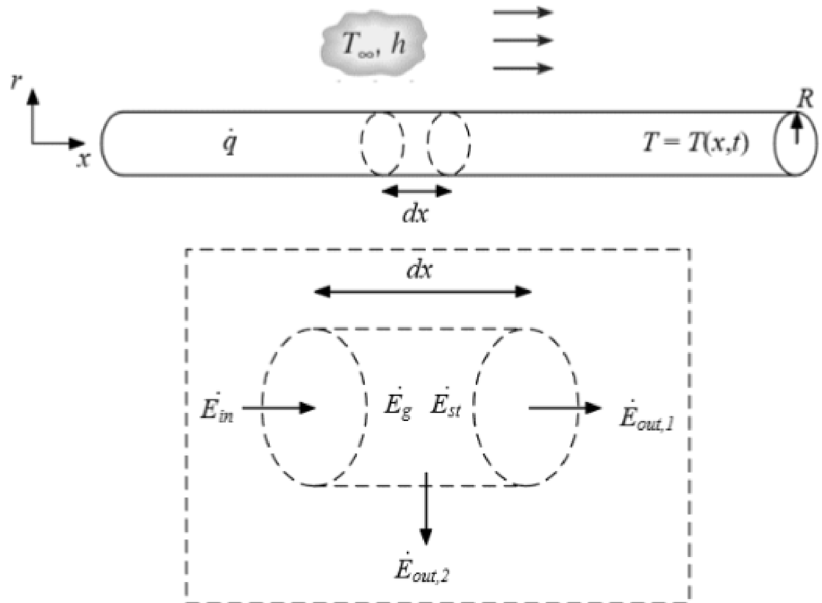
PROBLEM 1

KNOWN: Rod dimensions: R, L; environment: T_∞, h; volumetric heat generation rate: $\dot{q}$; $T(x = 0) = T_1$ for all t; $T(t = 0) = T_i$ for all x.

FIND: (a) For a differential cylinder, write down energy balance and express each term as a function of T; (b) Derive governing equation; (c) Express boundary conditions and initial conditions

SCHEMATIC:



ASSUMPTIONS: Very long rod, negligible thermal radiation, uniform radial temperature, i.e., $T = f(x, t)$, constant thermal and mechanical properties k, ρ, and c_p.

ANALYSIS: Part (a): The energy balance for a control volume is:

$$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \dot{E}_{st}$$
$$\dot{E}_{in} - (\dot{E}_{out,1} + \dot{E}_{out,2}) + \dot{E}_g = \dot{E}_{st}$$

Let $\Delta v \equiv A\mathrm{d}x$ where $A = \pi R^2$ and $\mathrm{d}x$ is the length of the differential control volume. Each term is:

$$\dot{E}_{in} = q_{cond} = \boxed{-kA\frac{\partial T}{\partial x}} \tag{1}$$

$$\dot{E}_{out,1} = q_{cond} = -kA\frac{\partial T}{\partial x} + \frac{\partial}{\partial x}\left(-kA\frac{\partial T}{\partial x}\right)\mathrm{d}x = \boxed{-kA\left(\frac{\partial T}{\partial x} + \frac{\partial^2 T}{\partial x^2}\mathrm{d}x\right)} \tag{2}$$

$$\dot{E}_{out,2} = q_{conv} = hP\mathrm{d}x(T_s - T_\infty) = \boxed{hP\mathrm{d}x(T - T_\infty)} \tag{3}$$

$$\dot{E}_g = \dot{q}\Delta v = \boxed{\dot{q}A\ \mathrm{d}x} \tag{4}$$

$$\dot{E}_{st} = \rho c_p\frac{\partial T}{\partial t}\Delta v = \boxed{\rho c_p\frac{\partial T}{\partial t}A\mathrm{d}x} \tag{5}$$

Part (b): Substitute the individual terms into the energy balance and simplify:

$$\dot{E}_{in} - \dot{E}_{out,1} - \dot{E}_{out,2} + \dot{E}_g = \dot{E}_{st}$$

$$\cancel{-kA\frac{\partial T}{\partial x}} - \left(-kA\left(\cancel{\frac{\partial T}{\partial x}} + \frac{\partial^2 T}{\partial x^2}\mathrm{d}x\right)\right) - (hP\mathrm{d}x(T - T_\infty)) + (\dot{q}A\mathrm{d}x) = \rho c_p \frac{\partial T}{\partial t}A\mathrm{d}x$$

$$k\frac{\partial^2 T}{\partial x^2}\cancel{\Delta v} - hP\mathrm{d}x(T - T_\infty) + \dot{q}\cancel{\Delta v} = \rho c_p \frac{\partial T}{\partial t}\cancel{\Delta v}$$

$$k\frac{\partial^2 T}{\partial x^2} - \frac{hP\mathrm{d}x}{\Delta v}(T - T_\infty) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{hP\mathrm{d}x}{\Delta v} = \frac{h \cdot 2\cancel{\pi}\cancel{R} \cdot \cancel{\mathrm{d}x}}{\cancel{\pi}R^{\cancel{2}}\cancel{\mathrm{d}x}} = \frac{2h}{R}$$

$$\boxed{k\frac{\partial^2 T}{\partial x^2} - \frac{2h}{R}(T - T_\infty) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}}$$

Part (c): We have a Dirichlet boundary condition at $x = 0$ and a Neumann boundary condition at $x = L$:

$$\boxed{T(x = 0, t) = T_1} \tag{6}$$

at left end: $q''_{conv} = q''_{cond} \implies -k\frac{\partial T}{\partial x} = h(T(x = L, t) - T_\infty) \implies \boxed{\left.\frac{\partial T}{\partial x}\right|_{x=L} = -\frac{h}{k}(T(L, t) - T_\infty)}$ (7)

The initial boundary condition can be expressed as $\boxed{T(x, t = 0) = T_i}$